Jane Street Puzzle October 2025

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The Artificial Automaton Athletics Association (Quad-A) is at it again, to compete with postseason baseball they are developing a Robot Baseball competition. Games are composed of a series of independent *at-bats* in which the batter is trying to maximize expected score and the pitcher is trying to minimize expected score.

An *at-bat* is a series of *pitches* with a running count of balls and strikes, both starting at zero. For each pitch, the pitcher decides whether to throw a ball or strike, and the batter decides whether to wait or swing; these decisions are made secretly and simultaneously. The results of these choices are as follows.

- If the pitcher throws a ball and the batter waits, the count of balls is incremented by 1.
- If the pitcher throws a strike and the batter waits, the count of strikes is incremented by 1.
- If the pitcher throws a ball and the batter swings, the count of strikes is incremented by 1.
- If the pitcher throws a strike and the batter swings, with probability p the batter hits a home run[1] and with probability $1-p$ the count of strikes is incremented by 1.

An at-bat ends when either:

- The count of balls reaches 4, in which case the batter receives 1 point.
- The count of strikes reaches 3, in which case the batter receives 0 points.
- The batter hits a home run, in which case the batter receives 4 points.

By varying the size of the strike zone, Quad-A can adjust the value p, the probability a pitched strike that is swung at results in a home run. They have found that viewers are most excited by at-bats that reach a *full count*, that is, the at-bats that reach the state of three balls and two strikes. Let q be the probability of at-bats reaching full count; q is dependent on p. Assume the batter and pitcher are both using optimal mixed strategies and Quad-A has chosen the p that maximizes q. Find this q, the maximal probability at-bats reach full count, to **ten decimal places**.

1. Robot fielding has a long way to go. ↩